



HOA SEN
UNIVERSITY

Lecture 3

SQL 2 – DML & Select Basic

Objectives

- INSERT, UPDATE and DELETE statements
 - Assertions and Triggers concept
 - Select Basic
 - Comparison operators
 - Between ... And, In
 - Like
 - Not – And – Or
 - More than one table query
-
- Reference: Chapter 6

Specifying Updates in SQL

- There are three SQL commands to modify the database:
 - **INSERT**
 - **DELETE**
 - **UPDATE**

INSERT

- In its simplest form, it is used to add one or more tuples to a relation
- Attribute values should be listed in the same order as the attributes were specified in the **CREATE TABLE** command

INSERT (2)

- Example:

```
U1:  INSERT INTO  EMPLOYEE
      VALUES ('Richard','K','Marini', '653298653',
              '30-DEC-52', '98 Oak Forest,Katy,TX', 'M',
              37000,'987654321', 4 )
```

- An alternate form of INSERT specifies explicitly the attribute names that correspond to the values in the new tuple
 - Attributes with NULL values can be left out

- Example: Insert a tuple for a new EMPLOYEE for whom we only know the FNAME, LNAME, and SSN attributes.

```
U1A: INSERT INTO EMPLOYEE (Fname, Lname, Ssn)
      VALUES ('Richard', 'Marini','653298653')
```

INSERT (3)

- Important Note: Only the constraints specified in the DDL commands are automatically enforced by the DBMS when updates are applied to the database
 - Another variation of INSERT allows insertion of *multiple tuples* resulting from a query into a relation

INSERT (4)

- Example: Suppose we want to create a temporary table that has the name, number of employees, and total salaries for each department.
 - A table DEPTS_INFO is created by U3A, and is loaded with the summary information retrieved from the database by the query in U3B.

```
U3A: CREATE TABLE  DEPTS INFO
      (Dept_name      VARCHAR(10) ,
       No_of_emps     INTEGER,
       Total_sal       INTEGER) ;
```

U3B:

```
INSERT INTO DEPTS_INFO (Dept_name, No_of_emps,
Total_sal)
SELECT      Dname, COUNT (*), SUM (Salary)
FROM        DEPARTMENT, EMPLOYEE
WHERE       Dnumber=Dno
GROUP BY    Dname ;
```

INSERT (5)

- Note: The DEPTS_INFO table may not be up-to-date if we change the tuples in either the DEPARTMENT or the EMPLOYEE relations *after* issuing U3B. We have to create a view (see later) to keep such a table up to date.

DELETE

- Removes tuples from a relation
 - Includes a WHERE-clause to select the tuples to be deleted
 - Referential integrity should be enforced
 - Tuples are deleted from only *one table* at a time (unless CASCADE is specified on a referential integrity constraint)
 - A missing WHERE-clause specifies that *all tuples* in the relation are to be deleted; the table then becomes an empty table
 - The number of tuples deleted depends on the number of tuples in the relation that satisfy the WHERE-clause

DELETE (2)

- Examples:

**U4A: DELETE FROM EMPLOYEE
WHERE Lname='Narayan'**

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DELETE (2)

- Examples:

**U4B: DELETE FROM EMPLOYEE
WHERE Ssn= '123456789'**

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DELETE (2)

- Examples:

**U4C: DELETE FROM EMPLOYEE
WHERE DNO IN
(SELECT Dnumber
FROM DEPARTMENT
WHERE Dname= 'Research')**

DEPARTMENT

Dname	<u>Dnumber</u>
Research	5
Administration	4
Headquarters	1

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DELETE (2)

- Examples:

U4D: DELETE FROM EMPLOYEE

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

UPDATE

- Used to modify attribute values of one or more selected tuples
- A WHERE-clause selects the tuples to be modified
- An additional SET-clause specifies the attributes to be modified and their new values
- Each command modifies tuples *in the same relation*
- Referential integrity should be enforced

UPDATE (2)

- Example: Change the location and controlling department number of project number 10 to 'Bellaire' and 5, respectively.

```
U5: UPDATE PROJECT
     SET      Plocation = 'Bellaire',
     Dnum = 5
     WHERE    Pnumber=10
```

PROJECT

Pname	<u>Pnumber</u>	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

UPDATE (3)

- Example: Give all employees in the 'Research' department a 10% raise in salary.

```
U6:    UPDATE    EMPLOYEE
        SET       Salary = Salary *1.1
        WHERE     Dno   IN (SELECT Dnumber
                             FROMDEPARTMENT
                             WHERE    Dname='Research' )
```

- In this request, the modified SALARY value depends on the original SALARY value in each tuple
 - The reference to the SALARY attribute on the right of = refers to the old SALARY value before modification
 - The reference to the SALARY attribute on the left of = refers to the new SALARY value after modification

Constraints as Assertions

- General constraints: constraints that do not fit in the basic SQL categories
- Mechanism: **CREATE ASSERTION**
 - Components include:
 - a constraint name,
 - followed by `CHECK`,
 - followed by a condition

Assertions: An Example

- “The salary of an employee must not be greater than the salary of the manager of the department that the employee works for”

```
CREAT ASSERTION SALARY_CONSTRAINT
CHECK (NOT EXISTS (SELECT *
                    FROM EMPLOYEE E, EMPLOYEE M,
                        DEPARTMENT D
                    WHERE E.Salary > M.salary AND
                        E.Dno=D.Number AND
                        D.Mgr_ssn=M.Ssn) )
```

Using General Assertions

- Specify a query that violates the condition; include inside a `NOT EXISTS` clause
- Query result must be empty
 - if the query result is not empty, the assertion has been violated

SQL Triggers

- Objective: to monitor a database and take initiate action when a condition occurs
- Triggers are expressed in a syntax similar to assertions and include the following:
 - Event
 - Such as an insert, deleted, or update operation
 - Condition
 - Action
 - To be taken when the condition is satisfied

SQL Triggers: An Example

- A trigger to compare an employee's salary to his/her supervisor during insert or update operations:

```
CREATE TRIGGER INFORM_SUPERVISOR
BEFORE INSERT OR UPDATE OF
  SALARY, SUPERVISOR_SSN ON EMPLOYEE
FOR EACH ROW
  WHEN
    (NEW.Salary > (SELECT Salary FROM EMPLOYEE
                  WHERE Ssn=NEW.Supervisor_ssn))
  INFORM_SUPERVISOR (NEW.Supervisor_ssn,NEW.Ssn) ;
```

Retrieval Queries in SQL

- SQL has one basic statement for retrieving information from a database; the **SELECT** statement
 - This is *not the same* as the SELECT operation of the relational algebra
- Important distinction between SQL and the formal relational model:
 - SQL allows a table (relation) to have two or more tuples that are identical in all their attribute values
 - Hence, an SQL relation (table) is a **multi-set** (sometimes called a **bag**) of tuples; it is *not* a set of tuples
- SQL relations can be constrained to be sets by specifying PRIMARY KEY or UNIQUE attributes, or by using the DISTINCT option in a query

Retrieval Queries in SQL (2)

- A **bag** or **multi-set** is like a set, but an element may appear more than once.
 - Example: $\{A, B, C, A\}$ is a bag. $\{A, B, C\}$ is also a bag that also is a set.
 - Bags also resemble lists, but the order is irrelevant in a bag.
- Example:
 - $\{A, B, A\} = \{B, A, A\}$ as bags
 - However, $[A, B, A]$ is not equal to $[B, A, A]$ as lists

Retrieval Queries in SQL (3)

- Basic form of the SQL SELECT statement is called a *mapping* or a SELECT-FROM-WHERE *block*

SELECT <attribute list>
FROM <table list>
WHERE <condition>

- <attribute list> is a list of attribute names whose values are to be retrieved by the query
- <table list> is a list of the relation names required to process the query
- <condition> is a conditional (Boolean) expression that identifies the tuples to be retrieved by the query

Retrieval Queries in SQL (4)

- Example of a simple query on one table

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
-------	-------	-------	------------	-------	---------	-----	--------	-----------	-----

Q1 : Retrieve the First name, Last name, birth date and address of the employee.

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

Retrieval Queries in SQL (5)

- Q2: List all Employee

Select *

From Employee

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

Where clause

- **Comparison operators:** =, !=, <>, <, <=, >, >=
- Q3: Retrieve the birthday and address of the employee whose first name is 'John'.

```
SELECT      Bdate, Address  
FROM        EMPLOYEE  
WHERE       Fname= ' John'
```

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

Where clause

- Q4 : Retrieve the last name, birth date and address of the employees whose last name is not 'Smith'.

```
SELECT      Lname, Bdate, Address  
FROM        EMPLOYEE  
WHERE       Lname <> 'Smith'
```

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

Where clause

- Q5: Retrieve the first name and last name of the employee whose dno is 5.

```
SELECT      Fname, Lname  
FROM        EMPLOYEE  
WHERE       Dno = 5
```

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

Where clause

- Q6: Retrieve the first name and last name of the employee whose salary is greater or equal 40000.

```
SELECT      Fname , Lname  
FROM        EMPLOYEE  
WHERE       Salary >= 40000
```

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

Where clause

- Q7: Retrieve the name, SSN and address of the employee whose birthdates 'Jul-31-1972'.

```
SELECT      Lname, Minit, LName, SSn  
FROM        EMPLOYEE  
WHERE       Bdate = '1972-07-31'
```

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate
John	B	Smith	123456789	1965-01-09
Franklin	T	Wong	333445555	1955-12-08
Alicia	J	Zelaya	999887777	1968-01-19
Jennifer	S	Wallace	987654321	1941-06-20
Ramesh	K	Narayan	666884444	1962-09-15
Joyce	A	English	453453453	1972-07-31
Ahmad	V	Jabbar	987987987	1969-03-29
James	E	Borg	888665555	1937-11-10

Where clause

- Q8: Retrieve the first name and last name of the employees who does not have his supper.

```
SELECT Fname, Minit, Lname
FROM      EMPLOYEE
WHERE      Supper_SSN is null
```

- Note that NULL indicates a value which is missing, not known, inappropriate, etc. NULL is not a blank or zero. NULL cannot be tested for equality with other NULL values.*

EMPLOYEE

Fname	Minit	Lname	Super_ssn
John	B	Smith	333445555
Franklin	T	Wong	888665555
Alicia	J	Zelaya	987654321
Jennifer	S	Wallace	888665555
Ramesh	K	Narayan	333445555
Joyce	A	English	333445555
Ahmad	V	Jabbar	987654321
James	E	Borg	NULL

Where clause

- **Between ... and ... operator**
- Q9: Retrieve the first name, last name and address of the employee whose birthdates June-01-1959 to Dec-31-1959.

```
SELECT Fname, Lname, address
FROM EMPLOYEE
WHERE Bdate Between '1959-06-01' and
'1959-12-31'
```

- Note that the BETWEEN predicate is inclusive. The above condition is equivalent to :

```
WHERE Bdate >= '1959-06-01' and Bdate <= '1959-
12-31'
```

Where clause

- In operator
- Q9B: Retrieve the first name, last name and address of the employee whose Dno is 1 or 4.

```
SELECT      Fname, Lname, address
FROM        EMPLOYEE
WHERE       Dno In (1,4)
```

- The above condition is equivalent to :

```
WHERE Dno = 1 or Dno = 4
```

Where clause

- Pattern matching
- Simple pattern matching is carried out using LIKE:

LIKE 'pattern-to-match'

- Where the pattern can include special wildcard characters:
 - % 0 or more arbitrary characters
 - _ any one character

- Query 10: Retrieve all employees whose address is in Houston, Texas. Here, the value of the ADDRESS attribute must contain the substring 'Houston,TX' in it.

Q10:

```
SELECT      Fname, Lname  
FROM EMPLOYEE  
WHERE Address LIKE '%Houston,TX%'
```

- Query 10B: Retrieve all employees who were born during the 1950s.
 - Here, '5' must be the 3rd character of the string (according to our format for date), so the BDATE value is '_____5_', with each underscore as a place holder for a single arbitrary character.

```
Q26:      SELECT Fname, Lname  
FROM EMPLOYEE  
WHERE Bdate LIKE '___5_____'
```

Where clause

- Not – And – Or operators

Q11: **Select ***

From Employee

Where Supper_SSn is Not Null;

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

Where clause

- Not – And – Or operators

Q12: **Select ***

From Employee

**Where Fname = 'Joyce' and Lname =
'English';**

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

Where clause

- Not – And – Or operators

Q13: **Select ***

From Employee

**Where Lname = 'English' or Fname =
'Jennifer';**

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

Distinct – Order clause

```
SELECT [DISTINCT] column_list  
FROM table_list  
    [WHERE condition]  
    [ORDER BY attribute[ DESC/ASC]  
        [,attribute [DESC,ASC]]...];
```

- Note that ASCending order is the default

```
Q14: SELECT FName  
FROM Employee  
WHERE Bdate < '1950-12-31'  
ORDER BY FName DESC;
```


- We can specify the keyword **DESC** if we want a descending order; the keyword **ASC** can be used to explicitly specify ascending order, even though it is the default
- Example:

```
SELECT  Dname, Lname, Fname, Pname  
        FROM DEPARTMENT, EMPLOYEE,  
        WORKS_ON, PROJECT  
        WHERE      Dnumber=Dno AND Ssn=Essn  
        AND Pno=Pnumber  
        ORDER BY Dname, Lname DESC
```

- Use Of DISTINCT
- SQL does not treat a relation as a set; duplicate tuples can appear
- To eliminate duplicate tuples in a query result, the keyword **DISTINCT** is used
- For example, the result of Q15 may have duplicate SALARY values whereas Q15A does not have any duplicate values

Q15: **SELECT Salary**
 FROM EMPLOYEE

Q15A: **SELECT DISTINCT Salary**
 FROM EMPLOYEE

Salary
30000
40000
25000
43000
38000
25000
25000
55000

Display Data from multiple tables

- Obtaining Data from Multiple Tables

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	...	Dno
John	B	Smith	123456789	...	5
Franklin	T	Wong	333445555	...	5
Alicia	J	Zelaya	999887777	...	4
Jennifer	S	Wallace	987654321	...	4
Ramesh	K	Narayan	666884444	...	5
Joyce	A	English	453453453	...	5
Ahmad	V	Jabbar	987987987	...	4
James	E	Borg	888665555	...	1

DEPARTMENT

<u>Dnumber</u>	Dname
5	Research
4	Administration
1	Headquarters

Fname	LName	SSN	Dno	Dnumber	Dname
Join	Smith	123456789	5	5	Research
...	...	...	...	...	...
James	Borg	888665555	1	1	Headquarters

- Syntax

```
SELECT      table1.column, table2.column  
FROM table1, table2  
WHERE       table1.column1 = table2.column2;
```

- *Use a join to query data from more than one table.*
- *Write the join condition in the WHERE clause.*
- *Prefix the column name with the table name when the same column name appears in more than one table.*

- Ex: Q16.

```
Select Fname, Lname, Dname  
From Employee, Department  
Wehre Employee.Dno = Department.Dnumber
```

Aliases

- Some queries need to refer to the same relation twice
 - In this case, *aliases* are given to the relation name
- Query 17: For each employee, retrieve the employee's name, and the name of his or her immediate supervisor.

```
Q17:      SELECT      E.Fname, E.Lname, S.Fname, S.Lname
           FROM        EMPLOYEE E S
           WHERE        E.Super_ssn=S.Ssn
```

- In Q17, the alternate relation names E and S are called *aliases* or *tuple variables* for the EMPLOYEE relation
- We can think of E and S as two different *copies* of EMPLOYEE; E represents employees in role of *supervisees* and S represents employees in role of *supervisors*

Aliases (2)

- Aliasing can also be used in any SQL query for convenience
- Can also use the AS keyword to specify aliases

**Q17B: SELECT E.Fname, E.Lname, S.Fname, S.Lname
FROM EMPLOYEE AS E, EMPLOYEE AS S
WHERE E.Super_ssn=S.Ssn**

EMPLOYEE E

Fname	Minit	Lname	<u>Ssn</u>	Super_ssn
John	B	Smith	123456789	333445555
Franklin	T	Wong	333445555	888665555
Alicia	J	Zelaya	999887777	987654321
Jennifer	S	Wallace	987654321	888665555
Ramesh	K	Narayan	666884444	333445555
Joyce	A	English	453453453	333445555
Ahmad	V	Jabbar	987987987	987654321
James	E	Borg	888665555	NULL

EMPLOYEE S

<u>Ssn</u>	Fname	Minit	Lname
123456789	John	B	Smith
333445555	Franklin	T	Wong
999887777	Alicia	J	Zelaya
987654321	Jennifer	S	Wallace
666884444	Ramesh	K	Narayan
453453453	Joyce	A	English
987987987	Ahmad	V	Jabbar
888665555	James	E	Borg

Joining More than Two Tables

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Dno
John	B	Smith	123456789	5
Franklin	T	Wong	333445555	5
Alicia	J	Zelaya	999887777	4
Jennifer	S	Wallace	987654321	4
Ramesh	K	Narayan	666884444	5
Joyce	A	English	453453453	5
Ahmad	V	Jabbar	987987987	4
James	E	Borg	888665555	1

DEPARTMENT

Dname	<u>Dnumber</u>
Research	5
Administration	4
Headquarters	1

DEPT_LOCATIONS

<u>Dnumber</u>	<u>Dlocation</u>
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

Fname	LName	SSN	Dno	Dnumber	Dname	Dnumber	Dlocation
Join	Smith	123456789	5	5	Research	5	Bellaire
Join	Smith	123456789	5	5	Research	5	Sugarland
Join	Smith	123456789	5	5	Research	5	Houston
...	...	...	...	...	...	...	...
James	Borg	888665555	1	1	Headquarter	1	Houston

UNSPECIFIED WHERE-clause

- Example:

Q18: **SELECT Ssn, Dname**
FROM EMPLOYEE, DEPARTMENT

- It is extremely important not to overlook specifying any selection and join conditions in the WHERE-clause; otherwise, incorrect and very large relations may result

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

How many rows?

- Query 19: For every project located in 'Stafford', list the project number, the controlling department number, and the department manager's last name, address, and birthdate.

```
Q19: SELECT      Pnumber, Dnum, Lname, Bdate, Address  
FROM      PROJECT, DEPARTMENT, EMPLOYEE  
WHERE      Dnum=Dnumber AND Mgr_ssn=Ssn  
AND Plocation='Stafford'
```

In Q19, there are two join conditions

- The join condition `Dnum=Dnumber` relates a project to its controlling department
- The join condition `Mgr_ssn=Ssn` relates the controlling department to the employee who manages that department

Arithmetic Operations

- The standard arithmetic operators '+', '-', '*', and '/' (for addition, subtraction, multiplication, and division, respectively) can be applied to numeric values in an SQL query result
- Q6: Show the effect of giving all employees who work on the 'ProductX' project a 10% raise.

```
SELECT      Fname, Lname, 1.1*Salary
FROM        EMPLOYEE, WORKS_ON, PROJECT
WHERE       Ssn=Essn AND Pno=Pnumber
            AND Pname='ProductX'
```

Q & A

